Paleomagnetic investigations in igneous–metamorphic rock units in eastern New England

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A clustering of paleomagnetic poles for rock units of mid-Paleozoic (ca. 400 Ma) age in eastern New England indicates that a variety of lithologic types may be used to obtain virtual geomagnetic poles (VGP's) in an igneous–metamorphic terrain. These rock types include chilled margins of basaltic dikes or narrow dikes, hydrothermal alterations such as epidote veinlets, mafic rocks reset by low-grade metamorphism, and hypabyssal igneous bodies, particularly where there is evidence of hydrothermal activity.

The results suggest the approach to be followed to obtain valid VGP's in similar geological belts in the Precambrian. The remanence directions in Devonian and Mississippian rock units in eastern Massachusetts and southern and northern Maine are close to other determinations of similar age within the Maritime block, demonstrating the potential usefulness of the procedures.

Un groupe de pôles paléomagnétiques d'unités lithostratigraphiques d'âge Paléozoïque moyen (environ 400 Ma) dans l'est de la Nouvelle-Angleterre indique qu'une variété de types lithologiques peut être utilisée pour obtenir les VGP's dans des terrains ignés et métamorphiques. Ces types de roches incluent les bordures de refroidissement des dykes basaltiques ou des dykes étroits, des veinules d'épidote ou autres altérations hydrothermales, des roches mafiques altérées par un faible métamorphisme, et des massifs de roches ignées hypobyssales, particulièrement où il y a indication d'une activité hydrothermale.

Les résultats suggèrent une méthode à suivre pour obtenir des VGP's valables dans des ceintures géologiquement analogues du Précambrien. Les directions de rémanence dans les unités lithostratigraphiques d'âges Dévonien et Mississipien dans l'est du Massachusetts, le sud et le nord du Maine sont très voisines des autres déterminations d'âge équivalent à l'intérieur du bloc des Maritimes, démontrant ainsi une utilisation potentielle de ces méthodes.

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Introduction

The principal objective of this project was to determine a reliable procedure for finding valid paleomagnetic pole positions in igneous–metamorphic terrains in the Precambrian. The plan was that a detailed study of a rising orogenic belt, such as the Appalachians, would provide a better opportunity for finding such methodology because of the much more detailed knowledge of superimposed events there, as compared with the case of the eroded equivalent in a Precambrian Shield area. The northern Appalachians, about which there has been a wealth of detailed geological and geochronological studies, was considered ideal. Metamorphic overprints are pervasive and are known regionally through K–Ar dating. Intrusive events have been dated by various methods so that in a number of specific cases their ages are known with some degree of confidence. Geological relationships and tectonic disturbances, which have taken place before the resetting of K–Ar age values, have been indicated to some extent by extensive field work so that resetting of paleomagnetic remanence should have been recognizable.

In accordance with this objective we have investigated the directions of paleomagnetic remanence in a wide variety of specific geological entities. These range in age from Late Precambrian granodioritic and mafic igneous complexes to the latest (340 Ma as determined by reset K–Ar values) hydrothermal alterations that appear to be later than all rotational tectonic disturbances. In an earlier report (Schutts *et al.* 1976) we outlined some successes in meeting this first objective, particularly in the use of rock systems that show pervasive alterations. Since the first report we have extended the investigation over rock systems from Massachusetts to New Brunswick, and from mafic igneous and metamorphic rocks to highly potassic granophyres. In general, the results have been successful, so that we now believe that it is possible to meet the principal objective if the investigator follows prescribed selective procedures.

General procedures

Magnetic data were obtained using a Schoenstedt spinner magnetometer and thermal demagnetizing system, and a design of alternating current (AC) demagnetizer by A. Brecher. The spinner was interfaced with a

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Hewlett-Packard model 9814A desk calculator and $X-Y$ plotter, and programmed for automatic plotting and printout of data. AC and thermal step-demagnetizations up to 1500 Oe (119 400 A/m) and 750°C on selected typical samples indicated that for the mafic igneous rocks with no obvious hematite the thermal demagnetization did not add significantly to the results. Most analyses were made, therefore, on the basis of AC demagnetization only. An average paleopole position for 24 specimens of Upper Triassic Palisade sill agreed well with the established pole position for that unit, and furnished an interlaboratory test of field and laboratory procedures (McElhinny 1973).

In plots of clustered points to follow, almost all of the measured data are included for each rock unit, except where an obvious geological reason (such as a granitic dikelet, or inclusion) explained the discrepant result. In our sampling procedure we started by taking many specimens per sample, but for the bulk of the work we took only two specimens per sample and increased the number of samples per site. Sampling, cutting, and measurement errors combined have a value of α_{95} less than 3°, as found in a cluster of points for separate samples taken from a single intrusive plug, described later.

Examples of AC and thermal demagnetization curves for high-quality specimens from a mafic complex are given in Fig. 1. The plots are presented as pole positions, and are typical of the reversed Devonian pole found for most of the cases studied. Specimens performing in this way were found in all of the areas investigated, giving credence to weaker cases where the characteristic magnetization was found by removal of viscous components.

Investigations related to the opaque mineralogy were reported earlier (Schutts *et al.* 1976). New work is described below, as it related to the rock units under consideration.

Results and discussion

A. Devonian and Mississippian

1. Mafic complex in Lexington, Belmont, Waltham, Arlington, Burlington, and Stoneham, Massachusetts

Bell and Dennen (1972) and Dennen (1976) have summarized the geological relationships between a number of igneous units in eastern Massachusetts described in the literature by investigators over a period of 70 years, and have brought them together under the name Plutonic Series in the Cape Ann area. The numerous units include the Salem gabbro–diorite, Cape Ann granite, and Peabody granite (Fig. 2). Geochronological data by Fairbairn *et al.* (1967), Fairbairn (1971), Handford *et al.* (1965), and Zartman and Marvin (1971) have pointed to an original age of 450 ± 15 Ma for the group as a whole. Field and petrographic study and compositional analysis indicate that the igneous units possess common characteristics, which vary continuously over the series.

Barosh *et al.* (1977) have compiled a geologic map of eastern Massachusetts in which they continue the Salem gabbro–diorite into a region northwest of Boston Basin in the towns of Lexington, Belmont, Waltham, Arlington, Burlington, and Stoneham. However, because of the considerable distance and changes in complexity of the rock systems we will refer to these units as the Mafic Complex northwest of Boston rather than the Salem gabbro–diorite. B. G. Naylor (personal communication, 1980) and others believe that these units represent a continuation of a Late Precambrian volcanic and sedimentary belt. The map in Fig. 2 follows boundaries taken from the compilation of Barosh *et al.* (1977). However, recent discoveries that probably place the Lynn and Mattapan volcanics in the Late Precambrian (Kaye and R. E. Zartman, unpublished data, 1980) and the Boston Basin in an age range earlier than Ordovician have prompted us to extend the Precambrian designations as shown in Fig. 2.

The Mafic Complex northwest of Boston is a hypabyssal association of rapidly varying gabbros, diabase, and basalt dikes, granite and aplite tongues and stringers, and hydrothermal alterations in no fixed sequence. The rocks are often laced with epidote veinlets giving them a stewed appearance in outcrop. Microscopically the rocks of the Mafic Complex are metagabbro, metadiabase, and somewhat less altered thin basalt dikes, where not veined. The pyroxenes are generally totally replaced by a uralitic hornblende, magnetite, epidote, biotite, or chlorite and the plagioclase is mostly replaced by saussurite, epidote, calcite, and sericite. The plagioclase is commonly in the range An_{24-45}; much of it has been albitized, with CaO going into epidote and calcite. The opaques are commonly distributed in close relationship with the alteration products. The rocks have often been invaded to a varying extent by granitic apophyses down in size to aplite stringers. The pervasive hydrothermal alteration involved introduction of large total proportions of H_2O and O_2 and produced a major metasomatism without shearing.

The dating of the hornblende and biotite in the rocks by total K–Ar analyses is given in Table 1. Presumably the difference in age (380 versus 340 Ma, respectively) reflects a cooling process and the blocking temperatures for Ar diffusion in the two minerals. If the Mafic Complex were an extension of the Salem gabbro–diorite its time of intrusion would be 450 ± 15 Ma (Zartman and Marvin 1971), but events in the Late Devonian and Mississippian have clearly modified it.

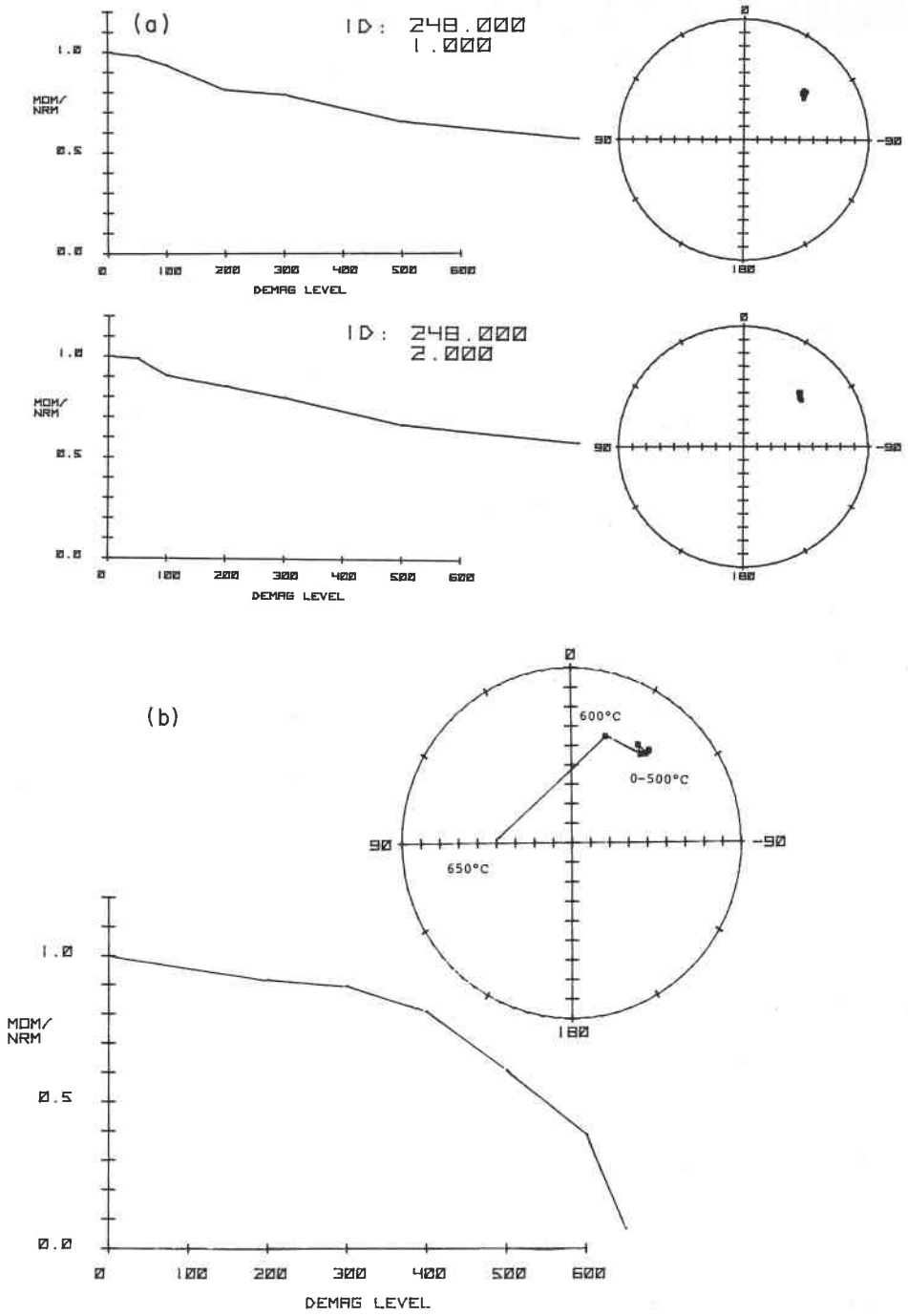


FIG. 1. (*a*) Plot of pole positions (north polar projection, longitude positive counterclockwise, points reversed in polarity) during AC demagnetization of two specimens cut from a single sample of hydrothermally altered chilled basalt dike. Location of points does not change during demagnetization owing to great stability of remanence. Units are in oersteds. (*b*) Similar plot of a specimen of aphanitic basalt, under thermal demagnetization.

Microscopic and probe analyses of the opaques in specimens from one of the best studied outcrops (Belmont Hill) showed two dominant Fe–Ti oxides and pyrite. In all specimens there was a magnetite with trellis exsolution lamellae of ilmenite. Grain sizes range from a maximum of 1 mm in coarser grained rocks down to a maximum of 0.02 mm in basaltic dike specimens; in addition there were much smaller grain sizes of dusty

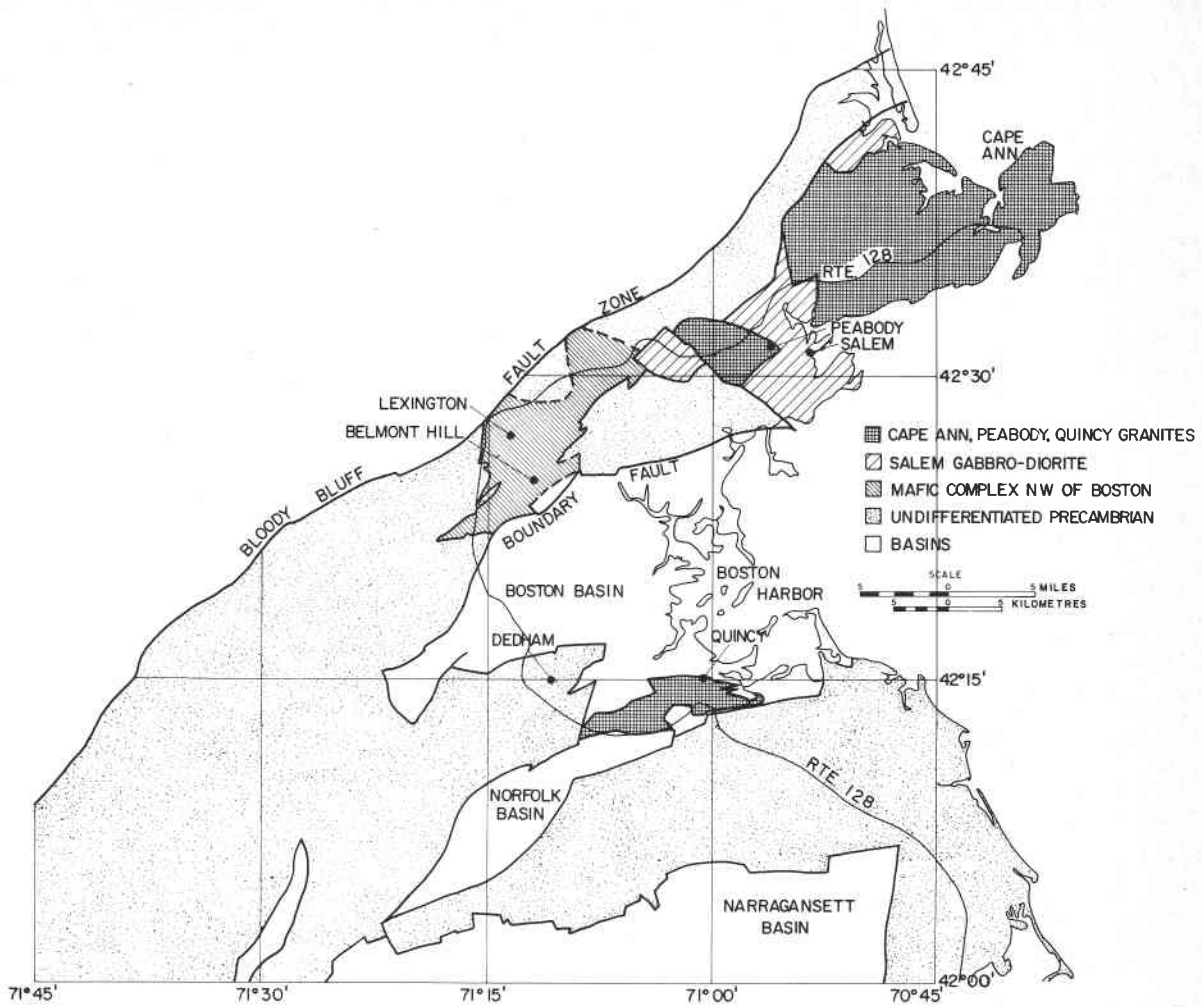


FIG. 2. Geological units sampled in eastern Massachusetts. Shown are geological boundaries of: Ordovician Plutonic Series of Cape Ann and Quincy granite; Precambrian or Ordovician Mafic Complex northwest of Boston, largely metamorphosed in the Late Devonian or Mississippian; Precambrian Dedham Complex and granites; and overlying basins (boundaries taken from Barosh *et al.* 1977).

magnetite in the alterations and groundmass. In general, the ilmenite lamellae, often 0.01 mm in width, enclose extremely small volumes of magnetite ($\leq 0.03\ mm^3$). Only rarely were large, subhedral, apparently primary, magnetite grains observed. These were partly resorbed, suggesting that small grains have been entirely consumed during metamorphic recrystallization.

An effort was made to use transmission electron microscopy (TEM) to determine the structure of the larger magnetite grains (Schutts *et al.* 1976). It was found that some of the larger grains were actually clusters of smaller grains, based on a differentiation of the orientations of their crystal structure, but we were not able to apply the technique to the ultimate extent of its capability, namely, to determine the domain state of the magnetite. A TEM study of the grains from a few of the samples indicates that the dusty magnetite grains are homogeneous and range in size down to 100 Å (10 nm). In the anhedral grains, the agglomerations consist of smaller particles ranging from 100 Å to several micrometres in size. Both the dusty component and the agglomerations, therefore, appear to contain superparamagnetic, single domain and multidomain grains. It is plausible (probable) that a large proportion of the stable remanence in the samples is carried by single domain like particles.

The oxides also occur as thick ilmenite lamellae (oxyexsolution) in or adjoining magnetite, or as ilmenite occurring independently. These differ from the ilmenite trellis lamellae in the magnetite, being lighter in color. The grain sizes of the ilmenite match those of the magnetite in the coarser grained rocks, and it becomes

TABLE 1. Radiometric age data

Location of rock unit	K–Ar mineral age (Ma) Biotite	Amphibole	Reference
Possible time of overprint by K–Ar age determination			
Mafic Complex from which paleomagnetic samples were taken:			
Trapelo Road, Lincoln	323	386	Schutts *et al.* (1976)
Route 2, Belmont Hill	359	380	Schutts *et al.* (1976)
Route 3, Lexington	346	342	Schutts *et al.* (1976)
Childs Street, Lexington	314	336	Schutts *et al.* (1976)
Belfrey Hill, Lexington	342	399	Schutts *et al.* (1976)
Granitoid contact rock near Ledgemont Laboratory, Lexington		377±6	R. E. Zartman (personal communication, 1980)
Alkaline granites	382	380	Zartman and Marvin (1971)
Cape Ann		393	Zartman and Marvin (1971)
		397	Zartman and Marvin (1971)
		374	Zartman and Marvin (1971)
		390	Zartman and Marvin (1971)
Peabody	367	370	Zartman and Marvin (1971)
		403	Zartman and Marvin (1971)
		384	Zartman and Marvin (1971)
		370	Zartman and Marvin (1971)
		350	Zartman and Marvin (1971)
		367	Zartman and Marvin (1971)
Average	340	380	

Rock unit	Age (Ma) Whole-rock Rb–Sr	Zircon Pb–U	Reference
Possible true ages of igneous events			
Younger invading plutons			
Cape Ann	435	450	Zartman and Marvin (1971)
Peabody	(367)	440	Zartman and Marvin (1971)
Quincy		440	Zartman and Marvin (1971)
Andover	460		Fairbairn *et al.* (1967)
Average	450		

less abundant in the finer grained rocks. In the basaltic dike there were separate ilmenite grains (0.02 mm), many of which were closely associated with sphene.

Probe analyses of these two oxides are given in Table 2. These data suggest a single dominant oxidation event, probably at the time of oxidation and hydration of the silicates, and at a low (submagmatic) temperature. The thermal demagnetizations point to an unblocking temperature of 500–600°C. This suggests that the carrier of the stable magnetization has a composition above Mt_{80}, as indicated in the analyses. The range of temperatures estimated for the development of the hornblende and lower grade metamorphism is from about 550°C down to about 300°C. If the hornblende Ar blocking temperature is about 500°C, the development of the secondary opaque mineralogy may have antedated the hornblende K–Ar ages. However, the necessity of the introduction of large amounts of H_2O, O_2, and K_2O during the alteration process suggests a deep geothermal circulation of supergene waters. This would cool the rock systems rapidly and telescope the time between hornblende and magnetite crystallization and argon retention by hornblende. We therefore place the time of the magnetic remanence in the Mafic Complex at 360 ± 20 Ma.

A combined plot of results from these six towns is given in Fig. 3. The means of the inclinations and declinations of the cleaned analyses are given in Table 3.

TABLE 2. Composition of co-existing Fe–Ti oxides in 55 probed cases

	Mean (%)	Standard deviation
Ilmenite		
TiO_2	49.26	0.78
FeO	48.21	0.91
MnO	2.24	0.25
Total	99.71	
Magnetite		
FeO	32.35	1.14
Fe_2O_3	65.45	3.46
TiO_2	1.15	1.23
Al_2O_3	0.55	0.34
Total	99.5	

NOTE: Range of TiO_2 in magnetite: 0.07–4.65%.

The natural remanent magnetization (NRM) data are shown in Fig. 3*a* and the final cleaned data in Fig. 3*b*. Three of the 53 specimens show normal polarity and the rest reversed.

A single road-cut exposure 100 m in length in Belmont, Massachusetts, was sampled in considerable detail. The outcrop shows at least four generations of cross-cutting events. Hypabyssal gabbros and diabases have been cut by apophyses of granite and aplite, basalt dikes, and epidote veinlets, not always in the same order. Within the experimental error all of the units sampled except for the granitic phases agree in orientation with the Mafic Complex as a whole, and have the same reversed polarity. The most stable and generally best specimens were epidote veinlets, chilled margins of narrow basalt dikes, and patchy areas formed with dusty magnetite from the hydrothermal alterations. In some of these samples the magnetic remanence did not change significantly in intensity or direction under AC demagnetizations up to 1500 Oe (119 400 A/m). The points are plotted in Fig. 4, and means for the cleaned results are given in Table 3.

One of the sites in Lexington represents a contact between Mafic Complex and intrusive granite. The granite has been dated by the U.S. Geological Survey using total K–Ar on hornblende, yielding an age of 377 ± 6 Ma (R. E. Zartman, personal communication, 1980). Samples include dark-colored host rocks and metasomatized host rocks with pink feldspar porphyroblasts. The plot of cleaned points is given in Fig. 5. The granite is probably a member of the Plutonic Series with an emplacement age of about 450 Ma. The site location is adjacent to the parking lot of the Ledgemont Laboratory of the Kennecott Copper Company on Spring Street, Lexington, Massachusetts.

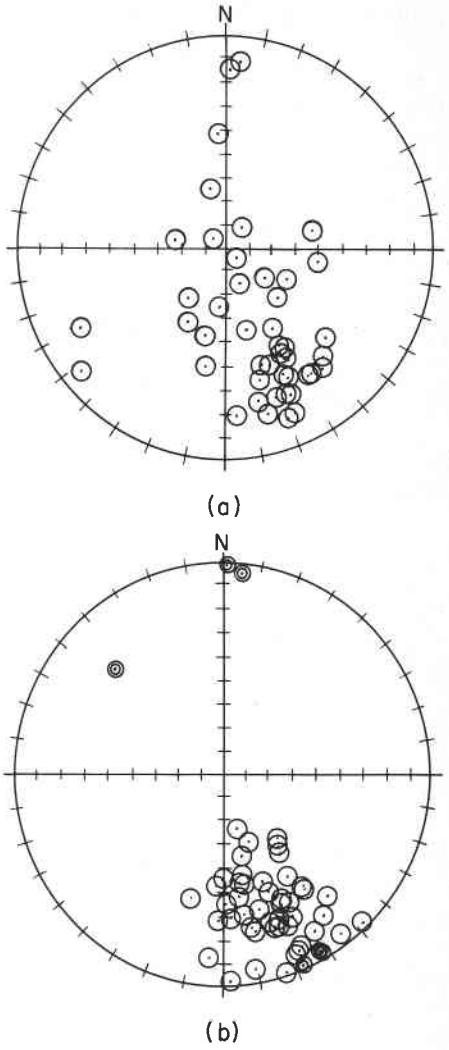


FIG. 3. Samples from Mafic Complex in Lexington, Belmont, Waltham, Arlington, Burlington, and Stoneham, Massachusetts. All hydrothermally altered gabbro, diabase, or basalt. (*a*) NRM and (*b*) AC cleaned data. In Figs. 3–9 the directions of remanence (*D*, *I*) are plotted with large open circles representing positive inclination and smaller double circles the reverse.

2. Possibly reset parts of the Salem gabbro–diorite

The typical Salem gabbro–diorite (Toulmin 1964) is dominantly a hornblende–augite, diorite, or gabbro having a fine to moderately coarse texture. Nearly all of the plagioclase (An_{40-50}) is highly altered to an aggregate of sericite, kaolinite, and calcite. The calcic cores of the zoned plagioclase crystals are more thoroughly altered than the more sodic rims. Hornblende is the principal dark mineral (up to 20%), and replaces augite almost totally in most specimens studied. Biotite is common, up to a few percent. Fe–Ti oxide minerals are ubiqui-

TABLE 3. Summary of pole positions indicated by the adopted cleaning procedures, and without statistical or other testing for possible multiple components

Location	N	$I°$	$D°$	$\alpha_{95}°$	K	P lat.	P long.	% reversed
				Mississippian and Devonian				
Mafic complexes								
Lexington *et al.*	53	24.5	161.7	5.6	18	−32.2	−50.0	94
Belmont	17	19.6	160.2	7.7	20	−34.6	−47.2	100
Contact zone	19	33.4	173.4	6.8	26	−29.0	−63.8	100
Salem A	29	39.5	169.4	9.4	10	−24.3	−60.1	92
NE Maine	12	34.6	170.3	16	11	−25.9	−58.0	75
Granites								
Webhannet	23	14	165	5.6	36	−38.2	−50.4	100
Red Beach	14	29.7	189.0	17	7	−28.6	−77.4	100
				Lower Cretaceous (?)				
Tatnic plug	18			2.9	136	61.5	−151.2	0

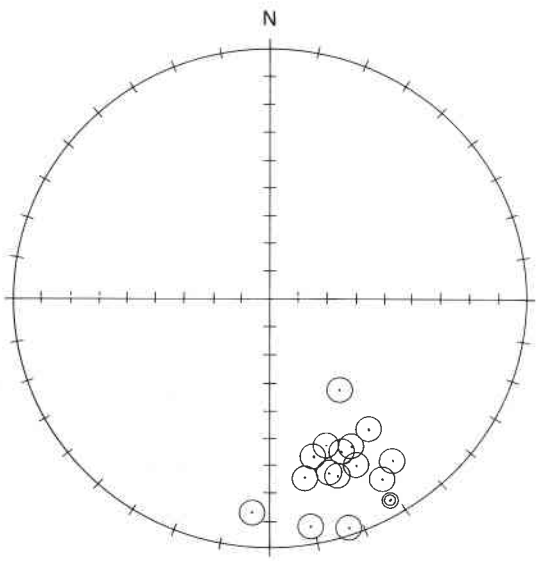


FIG. 4. Plot of remanence directions of AC cleaned data for samples from cross-cutting intrusives and hydrothermal alterations in a single exposure in Belmont, Massachusetts.

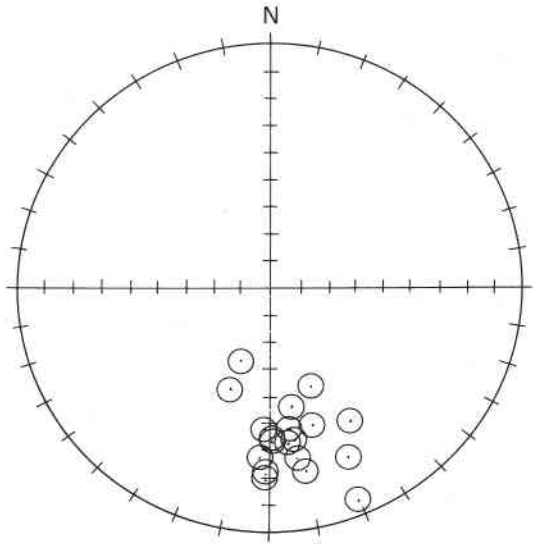


FIG. 5. Remanence directions of AC cleaned data points for samples of contact zone between intrusive granite and Mafic Complex, Lexington.

tous, and follow the description given generally for the Mafic Complex.

Basalt and diabase dikes cut the Salem gabbro–diorite in great abundance. Pegmatite dikes and pods, composed of pink feldspar, quartz, epidote, and more rarely calcite are common. Other cross-cutting intrusions are granite, micropegmatite, quartz porphyry, and similar rocks, and are clearly related to the granites of the Plutonic Series.

The paleomagnetic results in the Salem gabbro–diorite differentiate between northern (42.5°N, 70.8°W) and southern parts of the complex, with a dividing line trending approximately east–west and passing just north of the State Teachers College in Salem. The northern part has been designated Salem A, and shows a somewhat scattered paleoremanence generally in the same direction and reversed polarity as the Mafic Complex described above. NRM and AC cleaned plots are given in Fig. 6 with data for the means given in Table 3. Two cases of normal polarities support the direction of the characteristic magnetization.

There is a general similarity between the plots for the Mafic Complex in Lexington and surroundings (Fig. 3) and the plot of the Salem A group (Fig. 6). The Salem A group has a mean inclination of 39.5°, as opposed to 24.5° for the other group.

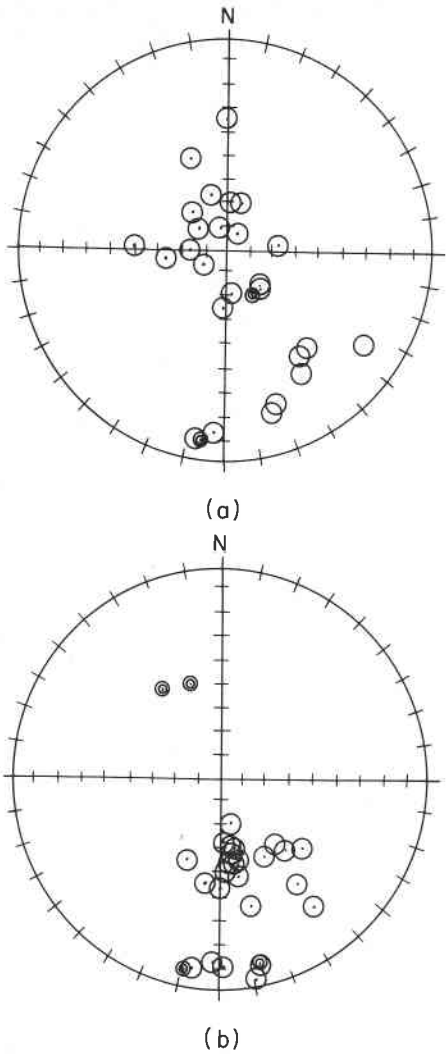


FIG. 6. Remanence directions for (*a*) NRM and (*b*) AC cleaned data for specimens of Salem A group of hydrothermally altered gabbro, diabase, and basalt.

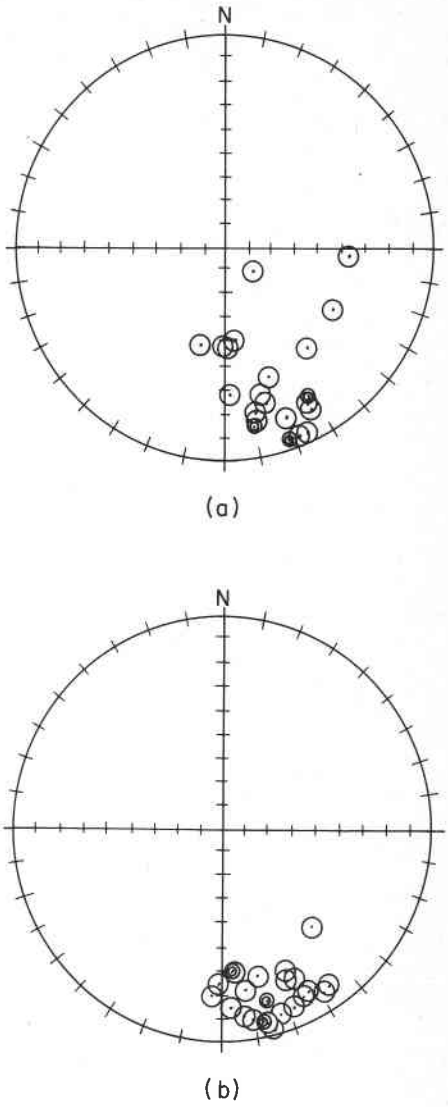


FIG. 7. Remanence directions for (*a*) NRM and (*b*) AC cleaned data for 23 specimens of granite and granodiorite from the eastern part of the Webhannet pluton, southern Maine (43.3°N, 70.7°W).

3. Webhannet pluton in southern Maine

This pluton in southern Maine (43.4°N, 70.7°W) has an age of 391 ± 42 Ma by whole-rock Rb–Sr analyses and 403 ± 2 Ma by Pb–U in zircons (Gaudette *et al.* 1977). The rock is typically a biotite granite with variations to quartz monzonite and granodiorite. It has regions in the eastern part of the pluton that showed strong, stable remanence (Fig. 7) and a clustering of points close to those of the Mafic Complex in Massachusetts. The polarity is 100% reversed, adding to the conclusion that we are approaching the true representative paleopole position for the Early to mid-Devonian in this block of continent. Means are given in Table 3.

4. Plutons in the Calais area, northern Maine

The plutons in this region next to the border with New Brunswick are shown in a map by Amos (1963). Units sampled include the granite of Baring, the Red Beach biotite granite, and the Red Beach hornblende granite, and an older gabbro and diorite. Rb–Sr whole-rock age data indicate an age of 393 ± 6 Ma for the Red Beach granites, and 413 ± 5 Ma for the Siluro-Devonian boundary (Spooner and Fairbairn 1970; Bottino and Fullagar 1966).

The granite of Baring and the Red Beach biotite granite showed weak and variable remanence, similar to findings on western locations in the Webhannet pluton and the Peabody granite in Massachusetts. Demagne-

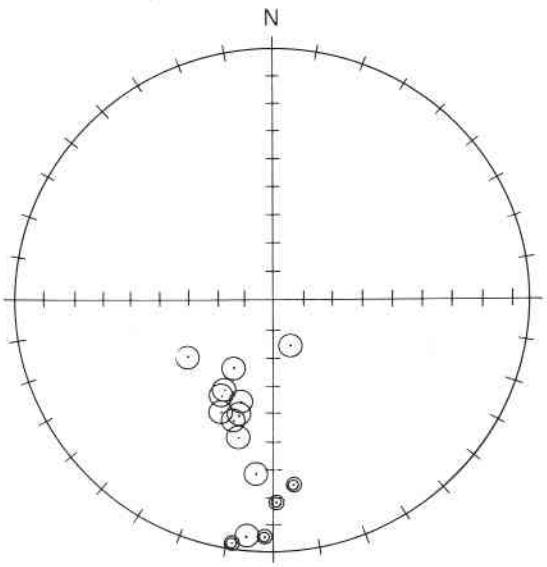


FIG. 8. Remanence directions for AC cleaned data for 16 specimens of hydrothermally altered Red Beach hornblende granite, extreme eastern Maine (45.1°N, 67.1°W).

tized results did not settle down well to an observable characteristic magnetization.

The so-called Red Beach hornblende granite (45.1°N, 67.1°W) shows a strong characteristic magnetization with little or no viscous component. The plot in Fig. 8 shows a pronounced streaking, which may be due to prolonged hydrothermal alteration superimposed on an original direction. The rock is a particularly interesting case because it is a granophyre made up mostly of a micrographic intergrowth of completely altered K-feldspar and clear quartz. Minor magnetite is associated with chlorite. There is no hornblende left and minor plagioclase is altered to epidote and clinozoisite. The rock is strongly reddish in color, probably due to hematite.

The Red Beach granites unconformably underlie the Devonian Perry Formation on which there has been an extensive paleomagnetic study. The data for the Perry Formation and other units of approximately similar age in the Maritime Provinces are assembled in a later figure.

5. Gabbroic complexes along the eastern coast of Maine

Several gabbroic complexes occur along the eastern coast of Maine within 30 km of the coastline and on Campobello Island, New Brunswick. A number of exposures are seen to consist mostly of diabasic and basaltic dikes or shallow gabbroic intrusives. The preliminary geologic map of Maine compiled under the direction of Doyle (1967) refers to these as gabbros and diorites of the Devonian New Hampshire Plutonic

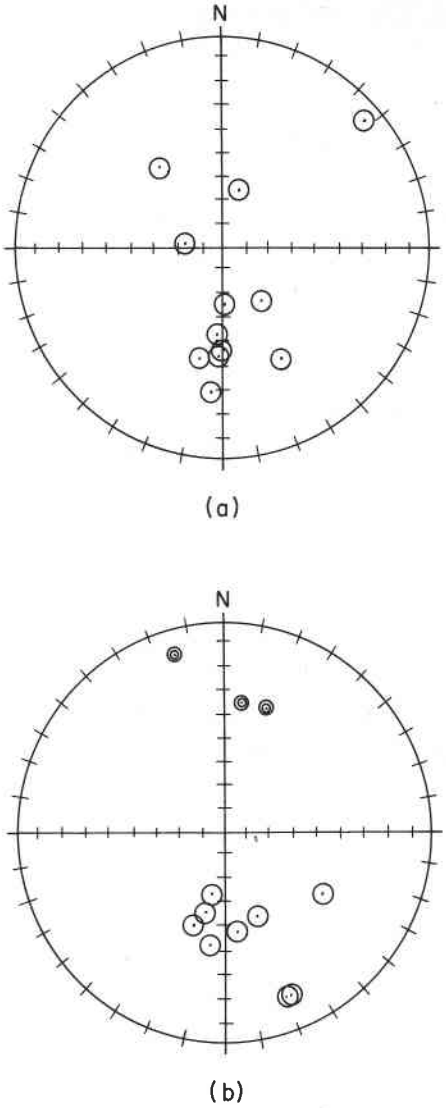


FIG. 9. Remanence directions for (*a*) NRM and (*b*) AC cleaned data for specimens from North Penobscot (44.48°N, 68.71°W), Sullivan (44.52°N, 68.14°W), and Jonesport (44.54°N, 67.62°W) areas, Maine, and Campobello Island (44.89°N, 66.94°W), New Brunswick.

Series. However, they are in a belt of stratified Silurian–Devonian rocks and may be of this age.

Paleomagnetic data on a sampling of the mafic masses close to the coast are shown in Fig. 9. Again, most of the directions are reversed; the normal polarities in three areas suggest an original early magnetization. The samples were taken from four localities: near North Penobscot (44.48°N, 68.71°W); near Sullivan (44.52°N, 68.14°W); near Jonesport (44.54°N, 67.62°W); and Campobello Island, N.B. (44.89°N, 66.94°W). Data for the results are given in Table 3.

6. Summary of Devonian and Mississippian pole positions

Figure 10 shows a comparison of the pole positions for rock systems in the age range 350–400 Ma as determined in this report, with poles for the Hopewell Group, New Brunswick (Roy and Park 1969, 1974), the Maringouin Formation, New Brunswick (Roy and Robertson 1968), and the Perry Formation, New Brunswick (Black 1964; Robertson *et al.* 1968). There is sufficient general agreement between the points to conclude that valid pole positions may be obtainable from metamorphosed and hydrothermally altered rock systems within continental mobile belts. In fact, the agreement is quite surprising in view of the considerable distance between the sampling regions and the fact that the ages are not precisely the same and the type of rock system is entirely different. The concordance of the results also suggests that the Avalon–Acadian (Maritime) block was not only continuous between New Brunswick and Massachusetts, but large enough to move without distortion in its fault motion along the eastern edge of North America.

Irving (1979) has compiled data on Devonian and Early Carboniferous paleopoles from stable North America, giving a mean of 48°N, 119°E ($N = 6$, $K = 130$, $\alpha_{95} = 3°$). This is considerably north and west of the positions shown in Fig. 10. Schutts *et al.* (1976), Kent and Opdyke (1978), and Irving (1979) have noted this discrepancy and concluded that the northeastern Appalachia–Maritime block moved into its present position relative to North America at some time later than the age of these paleopoles. It has been suggested that this block moved along a sinistral fault following the eastern edge of stable North America from a previous position that was about 14° of latitude further south than the North American continent in the Early Devonian, in order to accommodate this discrepancy. Recently Van der Voo (1979) has proposed that the Avalon plate, forming the eastern part of this Maritime block, was detached from Gondwana and accreted to North America in a converging plate motion that developed the Taconic Orogeny in the Ordovician. He further proposed that the Avalon plate was part of a larger assemblage involving parts of Britain and Europe, which he called the Amorica plate. If correct, this would imply that the regions in eastern Massachusetts shown as Precambrian and Ordovician in Fig. 2 moved rapidly northward from a South Polar location during the early Paleozoic.

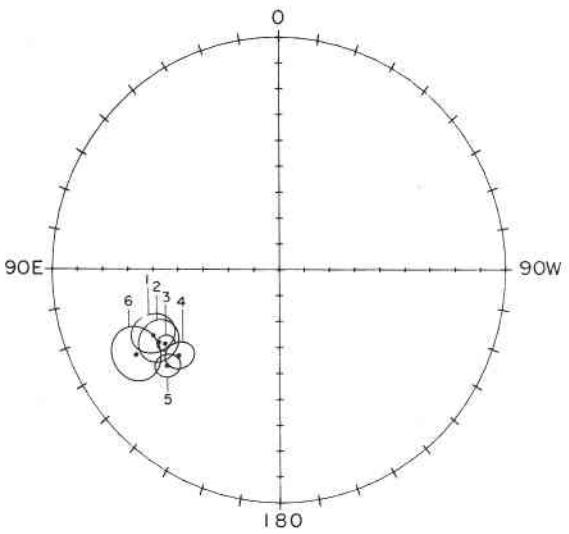


FIG. 10. North polar projection of equivalent normal poles with 95% confidence ovals determined from the means of measurements on Devonian or Mississippian rock units, compared with pole positions in the Maritime block of the same age range given in the literature. 1 = Maringouin; 2 = Perry; 3 = Hopewell; 4 = Webhannet; 5 = Lexington and other towns; 6 = Salem A. The confidence limits are given for the remanence directions found after uniform cleaning procedures without any attempt to define possible multiple components or otherwise improve the data.

7. Source of hydrothermal waters

The most significant part of this investigation is the demonstration that a post-tectonic, uniform, and almost complete metasomatism can affect large regions of mafic rocks and produce a stable overprint or remanence that obliterates any previous history. This remanence is so stable that in many cases it will not decrease much in intensity or change its direction under AF demagnetizing fields up to 1500 Oe (119 400 A/m). The metasomatism must be a commonplace occurrence because it has appeared almost everywhere in mafic rocks tested over distances of several hundred kilometres. In order to understand the cause of this alteration we attempted to determine whether it resulted from deep circulating meteoric waters heated by the last stages of intrusive activity, or whether the waters were hypergene, coming from deeply buried source regions within mafic volcanic complexes rich in pore fluids of marine origin. Another possibility might be that original marine geothermal systems caused hydrothermal alterations that liberated water upward during later metamorphism. Because the systems do not appear to have been rotated tectonically at any time after the overprint it does not appear likely that we are dealing with primary geothermal alterations that occurred during a submarine volcanic episode at the time of origin of the rocks.

Oxygen isotope ratios are most useful in demonstrating that a widespread alteration can be developed by late-stage geothermal systems. Following this possibility we obtained oxygen isotope determinations from H. P. Taylor of the California Institute of Technology on a collection of samples from the principal areas that showed prominent magnetic overprints. Most of these

samples showed petrographically an almost complete replacement of original minerals by hornblende, epidote, chlorite, or other alteration products. The oxygen isotope analyses were made on whole-rock samples, assuming a state of equilibrium between these fine-grained minerals. Table 4 lists Taylor's results.

Enough analyses were duplicated and showed an adequate reproducibility so that the data are acceptable. It will be noted that the granitic rocks have $\delta^{18}O$ values of 8.5–8.9 in the samples from Maine and the Belmont area, and 7.2–7.6 for the Cape Ann granite. The latter values are typical of many granitc rocks. The Red Beach granite is obviously altered to granophyre but there has not been any dramatic $\delta^{18}O$ change in it either. The $\delta^{18}O$ values of the mafic rocks vary from 5.0 to 6.5, except for the Maine gabbro that has $\delta^{18}O = 7.6$. Fresh tholeiitic basalt magma is fairly uniform at 5.8 ± 0.2, so that a few of these (notably 256A, 252A, 451B, 332A, and perhaps 249A) appear to have been slightly depleted in ^{18}O.

Unfortunately we cannot conclude whether or not the alterations in these rocks are caused by deep circulating meteoric waters in geothermal systems because at the equatorial location at that time the meteoric waters would not be sufficiently modified isotopically. An argument against the circulation of supergene waters is that many of the rocks show the effects of quick chilling of dike margins, and it is difficult to imagine a temperature of 300–400°C existing in these commonly hypabyssal systems some 40 Ma after their known time of origin.

This leaves the highly probable answer that the metasomatic hydration and oxidation were caused by hot ascending waters of metamorphic or marine connate origin. The circulating system must have been almost coextensive with the mafic rock systems as a whole because essentially no unaltered rocks have been found. The elevated temperature was brought up from depth by the circulating system, which must have been quite open as judged by the multiplicity of veins and monomineralic replacements, particularly by epidote. Whereas the mafic rock units at time of origin seem to have been under dilation, judging from the number of small dikes, the time of alteration might have coincided with a later Acadian compressional phase in which the pore fluids were generally moving upward.

We have concluded, therefore, that these mafic rock systems are typically water-rich piles of volcanic material, intruded by many dikes, with possible circulations of sea water during the volcanic stage, which subsequently redistributed the water in a pervasive, relatively high-temperature metasomatism of large volumes of the rock. This late-stage alteration was following any major tectonic rotation, and considerably later than the original time of volcanism.

TABLE 4. Oxygen isotope analyses by H. P. Taylor

Sample No.	Location	$\delta^{18}O$ analyses (‰)
Granites		
151A	Cape Ann	7.17
155B	Cape Ann	7.56
543A	Red Beach	8.58, 8.49
581B	Webhannet	8.97, 8.79
338B	Belmont	8.84, 8.98, 8.75
Gabbros, diorites, and basalts		
106A	Belmont	5.95, 6.19, 6.32
256A	Belmont	5.09, 4.95
332A	Belmont	5.28, 5.24, 5.12
249A	Lexington	5.23, 5.63
250A	Lexington	5.75
251A	Lexington	5.59, 5.63
252A	Lexington	5.25, 5.50
651A	Campobello Is.	5.80
652.2	Campobello Is.	5.73
658.2	Jonesport area	6.06
659.1	Jonesport area	6.09
499A	Calais area	6.23, 5.99, 5.98
521A	Calais area	6.52
451B	Swampscott	5.41, 5.30
453A	Marblehead	5.59
611.1	Sullivan area	7.58

B. A single Lower Cretaceous point in the Acadian–Avalon fault block

A circular mafic intrusion of gabbro laced with basaltic dikes occurs within the Webhannet pluton in southern Maine. It is known as the Tatnic plug, and probably is a member of the line of intrusions known as the Monteregian Hills, 120 ± 4 Ma in age.

All of the specimens of the gabbro and basalt showed extremely stable remanence with high intensities, which did not change much in value or direction under AC demagnetizations up to 1600 Oe (127 300 A/m). The polarities are all normal. The mean virtual geomagnetic pole (VGP) position is given in Table 3. The rocks are very fresh, and the fact that both host rocks and dikes give the same direction suggests that this is a valid VGP for this age in this block of crust. The pole position differs slightly from the average for the group of post-tectonic intrusives known as the Monteregian Hills, which have a mean paleopole at 72.4°N, 169.0°W (Foster and Symons 1979), but is within a possible secular variation if cooled rapidly. The points are plotted in Fig. 11. Taken as a true pole position, this case would suggest the possibility that the Acadian–Avalon fault block was several degrees more southerly in latitude at 120 Ma ago than the North American plate, extending the fault motion into much younger times than indicated by other evidence.

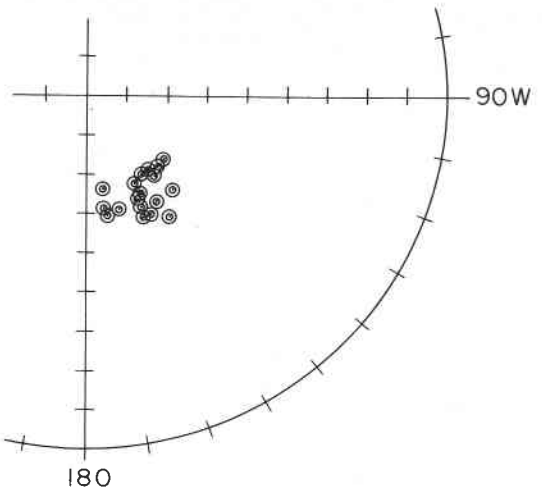


FIG. 11. Tatnic gabbroic intrusive in Webhannet pluton, 43.28°N, 70.70°W. Massive fresh medium-grained gabbro plug intruded by narrow, dense basalt dikes with margins chilled against the gabbro. Probably Early Cretaceous age. Highly stable remanence: all points taken after AC demagnetizations above 800 Oe (63 700 A/m), but little change prior to this level. Plot represents positive (normal) pole positions on north polar projection, longitude positive counterclockwise.

Conclusions

1. It is possible to obtain paleopole positions in igneous–metamorphic terrains if post-tectonic hydrothermal alterations are pervasive. The secondary magnetite may exhibit strong, stable remanence developed approximately at the time of Ar-blocking in the hydrous minerals.

2. Pole positions determined on such rock systems in eastern New England agree reasonably well with positions found in stratified rocks of the same age in New Brunswick, both testing the method and indicating the continuity of the Maritme block.

3. The paleopoles found for the age range 340–400 Ma yield further evidence for a more southerly location for the Maritime block, relative to stable North America, by about 14° of latitude.

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